

Expenses Reporting Tool for an IT Consultancy

Project Management (GEP)
Assignment 3: Budget and Sustainability Analysis

Bachelor Degree in Informatics Engineering
Specialization in Software Engineering

Bachelor's Thesis



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6 Budget and Sustainability

The final fundamental aspect for concluding the initial documentation of a project is the budget and the project's economic viability. In addition, it is important to analyse the project's sustainability, considering the impact of the proposed solution on economic, environmental, and social dimensions. This stage studies both positive and negative project consequences.

6.1 Budget

This section details all project expenses, justifying each cost and its corresponding calculation. The ultimate goal is to determine the initial project budget.

6.1.1 Personnel Cost

To obtain realistic personnel cost estimates, the average hourly wage for each personnel member was calculated in advance using market salary data [1]. This calculation assumes 260 working days per year with an average of 7.5 working hours per day.

In Table 6.1.1, the hourly wage calculation for each member of personnel is specified. Note that the final number already includes the Social Security (SS) cost. Also note that personnel are categorized by roles rather than by the human resources (HR) identified in Section 5.2.1. Each resource is assigned to at least one role in the table. There are five different roles involved in this project:

- **Project Manager (PM)**: Responsible for planning and leading the entire project.
- **Junior Full-Stack Developer (SE)**: Responsible for the full coding implementation of the application
- **QA Tester (T)**: Responsible for verifying the applications' functionality.
- **Software Architect (SA)**: Responsible for designing the application's technical structure while considering the project requirements and scope.
- **Technical Writer (TW)**: Responsible for elaborating a clear and accurate project documentation.

Role	Annual Salary (€)	Total including SS (€)	Cost / Hour (€)	Assigned HR
Project Manager	43,300	56,290	28.87	SM, PO, TD, GT
Full-stack Junior Developer	29,800	38,740	19.87	DT
QA Tester	26,888	34,954.4	17.93	DT
Software Architect	41,800	54,340	27.87	DT
Technical Writer	31,300	40,690	20.87	DT

Table 6.1.1: Annual and hourly salaries for different project roles. [Own creation]

In Table 6.1.2, personnel costs for each task and role, as well as the overall cost, are calculated. These costs result from multiplying each personnel member's effort hours (from Table 5.3.1) by their corresponding role's hourly wage (from Table 6.1.1).

ID	Name	Total hours	Hours per role					Cost (€)
			PM	D	T	SA	TW	
PS	Project Starting	50	80	0	0	0	0	1,433.33
PS1	Context and Scope	20	20	0	0	0	0	577.33
PS2	Temporal Planning	10	10	0	0	0	0	288.67
PS3	Budget and Sustainability	10	10	0	0	0	0	288.67
PS4	Initial Documentation	10	10	0	0	0	0	288.67
PM	Project Management	153	59	112	0	0	41	4,783.73
PM1	Agile Ceremonies	51	51	51	0	0	0	2,485.40
PM2	Thesis Director Review Meetings	8	8	8	0	0	0	389.87
PM3	Git Repository Documentation	40	0	40	0	0	0	794.67
PM4	User Manual Documentation	2	0	1	0	0	1	40.73
PM5	Product Backlog Update	8	0	8	0	0	0	158.93
PM6	Final Thesis Documentation	20	0	0	0	0	20	417.33
PM7	Final Demonstration Video	4	0	4	0	0	0	79.47
PM8	Bachelor's Thesis Defense	20	0	0	0	0	20	417.33
DT	Design	16	0	0	0	16	0	445.87
DT1	Use Case Diagram	2	0	0	0	2	0	55.73
DT2	Conceptual Model Diagram	4	0	0	0	4	0	111.47
DT3	Technology Stack Decision	2	0	0	0	2	0	55.73
DT4	Logic and Physic Architecture Design	4	0	0	0	4	0	111.47
DT5	UI mockups Design	2	0	0	0	2	0	55.73
DT6	Security Design	2	0	0	0	2	0	55.73
Dev	Development	36	0	31	5	0	0	705.49
Dev0	Self-learning and study of unfamiliar technologies	8	0	8	0	0	0	158.93
Dev1	Environment Configuration	2	0	2	0	0	0	39.73
Dev2	Product Deployment	8	0	8	0	0	0	158.93
Dev3	Unit and Integration Testing	10	0	5	5	0	0	188.96
Dev4	Logging and Monitoring	8	0	8	0	0	0	158.93
FBD	Frontend and Backend Development	128	0	128	0	0	0	2,542.93
FBD1	Expense Uploads	16	0	16	0	0	0	317.87
FBD2	Uploaded Expenses Real-time Dashboard	4	0	4	0	0	0	79.47
FBD3	AI-Extracted Expense Data Review and Correction	20	0	20	0	0	0	397.33
FBD4	Uploaded Expenses Deletion	4	0	4	0	0	0	79.47
FBD5	Expense Reports View, Creation, Edit, Submission, and Deletion	20	0	20	0	0	0	397.33
FBD6	Company Email User Authentication	8	0	8	0	0	0	158.93
FBD7	Role-based User Access	16	0	16	0	0	0	317.87

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ID	Name	Total hours	Hours per role					Cost (€)
			PM	D	T	SA	TW	
FBD8	Submitted Report Approval and Rejection	8	0	8	0	0	0	158.93
FBD9	Employee Spending Limit Configuration	8	0	8	0	0	0	158.93
FBD10	Submitted and Reviewed Expenses Record Visualization	4	0	4	0	0	0	79.47
FBD11	Email Notification Service Integration	20	0	20	0	0	0	397.33
BD	Backend Development	120	0	120	0	0	0	2,384.00
BD1	RESTful API Backend Logic Implementation	20	0	20	0	0	0	397.33
BD2	Azure SQL Database Integration and Manipulation	20	0	20	0	0	0	397.33
BD3	Azure Blob Storage Integration	8	0	8	0	0	0	158.93
BD4	Microsoft Document Intelligence AI Model Integration	20	0	20	0	0	0	397.33
BD5	Automated AI-driven Data Extraction Pipeline	20	0	20	0	0	0	397.33
BD6	Azure SignalR Real-time Notification Service Integration	16	0	16	0	0	0	317.87
BD7	Microsoft Entra ID User Authentication Integration	16	0	16	0	0	0	317.87
FD	Frontend Development	10	0	10	0	0	0	198.67
FD1	Expenses Search and Filter	8	0	8	0	0	0	158.93
FD2	Reports Search	2	0	2	0	0	0	39.73
Total		513	109	401	5	16	41	12,504.03

Table 6.1.2: Personnel cost breakdown by task, role, and total personnel cost. [Own creation]

6.1.2 Generic Cost

In addition to personnel costs, other project costs must also be considered. Therefore, the missing costs are detailed in this section, followed by the formula applied to obtain the resulting cost.

Amortization

Physical resources are also required and used in the project. However, considering only the product purchase price would not be entirely correct. For this reason, the depreciation of each hardware resource has been calculated.

Hardware resources have a 4-year depreciation period. Other relevant factors for amortization calculation include the total project duration and the number of hours dedicated to the project each day, respectively 122 days and 4 hours per day. Since the hardware resources are used throughout the project timeline, their total usage time corresponds to the total project hours (513 hours). Accordingly, the following equation calculates the amortization costs for each hardware resource shown in Table 6.1.3:

$$\text{Amortization (€)} = \text{Resource price} \times \frac{1}{4 \text{ years}} \times \frac{1}{122 \text{ days}} \times \frac{1}{4 \text{ hours}} \times \text{hours used}$$

Hardware Resources	Price (€)	Quantity	Total Price (€)	Usage Time (h)	Amortization (€)
PC Dell 16 Pro [2]	1,193.53	1	1,193.53	513	313.67
Wireless Mouse [3]	11.99	1	16.99	513	4.47
Dell Monitor [4]	171.82	2	343.64	513	90.31
Keyboard [5]	19.99	1	19.99	513	5.25
Total					408.44

Table 6.1.3: Amortization cost of the project's hardware resources. [Own creation]

Electric cost

The current cost of electricity is around **0.1748 €/kWh** [6]. Electricity costs are calculated based solely on the electricity-consuming hardware resources and their respective usage times. Following the same approach used for amortization cost calculation, the usage time is **513 hours**. Table 6.1.4 details the calculations and the resulting total cost:

Hardware Resources	Power (W)	Usage Time (h)	Energy Consumed (kWh)	Electricity Cost (€)
PC Dell 16 Pro [2]	65	513	33.345	5.83
Wireless Mouse [3]	0.05	513	0.02565	0.00
Dell Monitor [4]	24	513	12.312	2.15
Keyboard [5]	0.1	513	0.0513	0.01
Total				7.99

Table 6.1.4: Electricity cost of the project's hardware resources. [Own creation]

Internet cost

The basic internet plan costs €21.95 per month [7]. Since the project lasts 122 days, the internet cost is calculated as follows:

$$\text{Internet cost} = 21.95 \text{ €/month} \times \frac{1 \text{ month}}{30 \text{ days}} \times 122 \text{ days} = \mathbf{€89.26}$$

Travel cost

The public transport rate in Barcelona is 45.5€ per 3 months [8]; the total travel cost corresponds to the following equation:

$$\text{Travel cost} = 45.5 \text{ €/3 months} \times \frac{3 \text{ months}}{90 \text{ days}} \times 122 \text{ days} = \mathbf{€61.88}$$

Office rental cost

The project development always takes place in the FSP office. Therefore, this is the only cost to consider in this category. The FSP office is located in Plaça Francesc Macià, Barcelona. The monthly rental fee is €6,000. Given that there is enough space to accommodate up to 25 employees working at the same time, the total cost is calculated as follows:

$$\text{Office rental cost} = 6,000 \text{ €/month} \times \frac{1 \text{ month}}{30 \text{ days}} \times \frac{1 \text{ office}}{25 \text{ employees}} \times 122 \text{ days} = \mathbf{€976.00}$$

Software resources cost

Finally, the cost of software resources is also considered. However, most of them are open-source and therefore excluded from the total cost, as they are available for free. Only resources that require payment are listed below. Regarding Azure resources, only those that incur a cost are included, since Azure offers a pay-as-you-go fee [9]. See Table 6.1.5 for a detailed breakdown of the cost of Azure resources.

Azure resources	Cost (€/month)
SQL DataBase	3.90
Bandwidth	0.01
Blob Storage	0.01
Log Analytics	0.01
Event Grid	0.01
Service Bus	0.01
Functions	0.01
Total	3.96

Table 6.1.5: Azure resource cost. [Own creation]

Given that the Azure resource cost is €3.96 per month, the total cost for the project (122 days) is calculated as follows:

$$\text{Azure resource cost} = 3.96 \text{ €/month} \times \frac{1 \text{ month}}{30 \text{ days}} \times 122 \text{ days} = \mathbf{€24.16}$$

Microsoft is also included in the software resources cost, demanding an annual fee of €99 [10]. This resource cost is calculated in the following equation:

$$\text{Microsoft resource cost} = 99 \text{ €/year} \times \frac{1 \text{ year}}{365 \text{ days}} \times 122 \text{ days} = \mathbf{€33.09}$$

Thus, the total software resources cost equals the sum of both costs:

$$€24.16 + €33.09 = \mathbf{€57.25}$$

Total generic cost

By summing all previously calculated costs, the total generic cost is obtained, as shown in Table 6.1.6.

Concept	Cost (€)
Amortization	408.44
Electricity	7.99
Internet	89.26
Travel	61.68
Office Rental	976.00
Software	57.25
Generic Cost (GC)	1,600.63

Table 6.1.6: Total generic cost. [Own creation]

6.1.3 Other costs

Contingencies

No project strategy can eliminate the possibility of encountering obstacles. Therefore, a time buffer (27 hours) has been included to cover potential setbacks. However, simply adding extra time is insufficient, as it also entails additional resource consumption. Consequently, a portion of the project budget must be allocated for these potential scenarios. The proportional buffer time allocated to the total project duration is 5% (27h/420). Therefore, the contingency cost amounts to **€705.23**, which is the 5% of the total general cost calculated so far (PC (€) + GC (€) = 13,370.03 + 1,600.92 = €14,970.94).

Incidental Costs

Incidental costs encompass the project risks outlined in Section 5.5. The cost of each risk is calculated by considering its probability, time cost, and required resources. Table 6.1.7 details these calculations. Note that the resources required for each risk are identified as Rx (R1, R2, ...).

Incident	Risk (%)	Time (h)	R1	R1 Cost (€/h)	R2	R2 Cost (€/h)	Cost (€)	Risk Cost (€)
Coding Bugs	30	10	DT	19.87	HW	7.96	206.63	61.99
Tight Project Deadlines	15	20	PM	28.87	HW	15.92	593.26	88.99
Lack of Experience in Certain Technologies	30	8	DT	19.87	HW	6.37	165.30	49.59
Company's Licenses and Security Permissions	15	0	DT	19.87	HW	0.00	0.00	0.00
Total								200.57

Table 6.1.7: Incidental cost. [Own creation]

6.1.4 Total cost

After summing the total costs from each section, the final project budget is determined, as shown in Table 6.1.8.

Concept	Cost (€)
Personnel Cost (PC)	12,504.03
Generic Cost (GC)	1,600.63
Contingency Cost (CC)	705.23
Incidental Cost (IC)	200.57
Total Budget	15,010.45

Table 6.1.8: Total cost of the project. [Own creation]

6.1.5 Management Control

Challenges, both expected and unexpected, are always present in any project. Even with a precisely estimated timeline and budget, it is very common for a project to fall short of the initial plan. To monitor costs during project execution, various deviation formulas will be applied in the following areas:

- **Human resources deviation:** To contrast the actual effort with the previous estimated effort of human resources in each task.

$$\text{Human resources deviation} = \sum_{i \in pit} (\text{estimated_cost_per_hour}_i - \text{real_cost_per_hour}_i) \times \text{total_real_hours}_i$$

Where i indexes each individual in pit , the set of all personnel involved in that task.

- **Amortization deviation:** To compare the previously estimated use of hardware resources with the actual usage time of those resources.

$$\text{Amortization deviation} = \sum_{i \in hr} (\text{estimated_usage_hours}_i - \text{real_usage_hours}_i) \times \text{price_per_hour}_i$$

Where i indexes each resource in hr , the set of all hardware resources used in the project.

- **Total cost deviation:** To determine each task's real cost deviation from the estimated cost. The general cost in the following formula represents the sum of the personnel and generic costs, excluding contingency and incident costs. Hence, the following formula is applied to each task:

$$\text{Total cost deviation} = (\text{estimated_general_cost} - \text{real_general_cost})$$

All cost deviation calculations require the real project costs across different categories. Before applying the deviation formulas, the real costs are calculated at the end of the project using the same methodology as the initial budget estimation.

Note that the resulting total cost deviation value can be either positive or negative. A positive value indicates an overestimation of the initial budget for that resource or task. In this case, the additional budget can be reallocated to address issues that arise during the project. Conversely, a negative value indicates an underestimation, requiring contingency and incident costs to cover the budget shortfall.

6.2 Sustainability

The technological revolution has accelerated dramatically over the last few decades. A clear example is the launch of the first telephone (1876), the first computer (1937), the first laptop (1974), the first smartphone (2007), and the first artificial intelligence (2017) [11]. This trend reveals that the time it takes for a technological invention to emerge and significantly impact society continues to shrink. This exponential growth underscores the increasing importance of evaluating the economic, environmental, and social impacts of technological projects. These three dimensions together constitute sustainability, the topic discussed in this section.

Technology is shaping our future, at least in the coming years. The primary mission of every engineer is to make people's daily lives easier. However, as mentioned earlier, developing a solution is not enough; a study of its sustainability effects is also a must. Therefore, addressing sustainability questions enables a better evaluation of the project's impact across the economy, environment, and society. This analysis raises awareness of the weight of sustainability in project development.

The author's sustainability consciousness is quite limited. The knowledge extends only to well-known aspects such as environmental footprint and social impact. The economic dimension and multiple sustainability indicators require further study. The following section answers a series of key sustainability questions within the author's current expertise.

6.2.1 Economic Impact

Regarding PPP: Have you estimated the project execution cost?

The project budget is detailed in Section 6.1. All required costs have been considered, including personnel, hardware, software, contingencies, and incidents. Only indispensable resources are used and are included in the budget.

The computer used throughout the project was recently purchased and remains under warranty. Since there is an overabundance of hardware resources in the office, they can be easily replaced if needed. Regarding the software resources, they were already available to the company and are virtual; as a consequence, fixes and repairs are not included in the budget.

Additionally, only essential tasks demand a higher time cost. Since the AI model is connected to an external resource, this development phase can be significantly reduced while obtaining accurate data extraction. From an objective standpoint, a meticulous budget estimate has been established.

Regarding Useful Life: How are currently solved economic issues (costs...) related to the problem that you want to address (state of the art)?

The current expense management process imposes an excessive time cost for employees, who manually enter expenses into Excel spreadsheets. The same challenge is encountered in the finance department, which must verify the accuracy of each line item before approving reimbursements. Additionally, the current process is prone to human error regarding manual expense data entry and review.

In short, this time-consuming, low-efficiency process delivers minimal value while misallocating company personnel costs for both employees and finance staff.

Regarding Useful Life: How will your solution improve economic issues (costs, etc.) with respect to other existing solutions?

The proposed solution automates the current cumbersome process. It provides automatic receipt data extraction and a more intuitive dashboard for displaying receipt information. This eliminates the tedious manual data entry, enhances visualization beyond traditional spreadsheets, and reduces human error. As a result, the workflow is significantly streamlined, delivering accurate receipt processing while maximizing efficiency. This enables employees to focus on higher-value tasks, thus maximizing personnel investment returns.

6.2.2 Environmental Impact

Regarding PPP: Have you estimated the project's environmental impact?

The only resource directly affecting the environmental footprint is energy consumption. This resource is required throughout the entire project, as specified in Section 6.1.2. Both hardware resources and the shared workspace consume electricity continuously. However, hardware resource consumption is vital for application development and sharing the workspace's lighting with other employees reduces its environmental impact.

Regarding PPP: Have you considered minimizing its impact, for example, by reusing resources?

The application uses Azure for software resources. Currently, the Azure resources required for application development cost less than traditional physical resources. This cost optimization results from significantly reduced resource consumption, as cloud resources eliminate the need for transportation and frequent physical replacements. Updates and fixes are performed virtually, further reducing hardware requirements and electronic waste.

Furthermore, training the AI model requires substantial energy consumption. Reusing existing solutions, as decided in this project, optimizes this energy-intensive development process. Hence, the entire application development process prevents excessive resource use. As a result, tasks can be significantly reduced while avoiding excessive time costs.

Regarding Useful Life: How is the problem that you want to address currently solved (state of the art)?

The extensive time employees and finance staff dedicate to Excel spreadsheets results in excessive energy use. Switching the personnel to process expenses in the application also requires an energy-saving approach.

The primary strategy for reducing resource consumption is to develop the most efficient application possible, avoiding unnecessary resources and inefficient code that could overuse resources. This ensures that only essential resources are consumed each time employees use the application, limiting overall resource consumption throughout its lifecycle. Therefore, writing high-quality code is an essential skill for a programmer and is strongly considered throughout the project development.

Regarding Useful Life: How will your solution improve the environment with respect to other existing solutions?

This application is tailored for the company's exclusive use and excludes any irrelevant functionality from the final product. It streamlines resource consumption by avoiding the unnecessary computation performed by other existing applications.

6.2.3 Social Impact

Regarding PPP: What do you think you will achieve—in terms of personal growth—from doing this project?

This project enables the author to apply prior academic knowledge and software development skills in a real-world context. As a result, it enriches their working experience as future software engineer. Furthermore, it is the author's first experience managing and leading an entire project independently, strengthening self-learning, self-management skills and the ability to build software projects from scratch.

Regarding Useful Life: How is the problem that you want to address currently solved (state of the art)?

The current expense management process is handled manually through traditional methods. This involves manually entering data into Excel spreadsheets, emailing them to the finance department, and reviewing them line by line.

Regarding Useful Life: How will your solution improve the quality of life (social dimension) with respect to other existing solutions?

The proposed solution optimizes time spent on expense management while reducing employees' frustration and fatigue. This frees employees from this tedious task, allowing additional time to focus on more meaningful work. As a result, it maximizes employees' productivity.

Regarding Useful Life: Is there a real need for the project?

Regarding all project sustainability impacts (social, environmental, and economic), there is a clear need to build a robust solution. This solution overcomes challenges from all involved parties while introducing a sustainability-focused approach.

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